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Source: *Conflict Management and Peace Science*, April 2011, Vol. 28, No. 2 (April 2011), pp. 111-123

Published by: Sage Publications, Ltd.

Stable URL: https://www.jstor.org/stable/26275306

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[DOI:10.1177/0738894210396631]
Vol 28(2): 111–123

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Unpacking Escalation*

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Past studies vary widely in how they measure conflict escalation. The use of different measures of escalation is understandable because theories about escalation do not specify how the concept should be measured. But the use of different measures of escalation hinders cumulation. We contribute to the literature on conflict escalation by comparing a variety of measures of escalation. We demonstrate how few robust relationships there are across different measures of escalation. Importantly, though, we find a consistent influence of territory on escalatory processes which is insensitive to how escalation is measured.

KEYWORDS: concept measurement; conflict escalation; Militarized Interstate Disputes (MIDs); selection effects; territory

The escalation of hostilities is one of the most prominent concepts in the analysis of international conflict. It is the explicit behavior of interest in many theoretical studies of conflict and is the dependent variable in many quantitative tests of hypotheses about such behaviors. However, very few investigations permit their causes to vary across different types of escalation. In both theoretical and empirical work, "escalation" is usually modeled as a non-controversial single phenomenon. We believe that significant progress in understanding escalation will only be achieved after careful description of (and subsequent theorizing about) both sources and types of escalatory behavior. To enable future work on such matters, we offer here a broad analysis of escalatory patterns in which we determine how robust relationships are between alleged correlates and outcomes across different indicators of escalation.

We begin by summarizing existing research on conflict escalation. We next describe six measures of conflict escalation and undertake analyses to determine whether relationships asserted by past researchers persist across these six measures of escalation. We conclude by describing what our findings suggest for further theory building about and testing of models of escalation.

* For comments on and reactions to earlier versions of this article we are grateful to the anonymous reviewers and to Scott Bennett, Bruce Bueno de Mesquita, Harvey Starr, Jakana Thomas, John Vasquez, Chris Zorn, and, particularly and especially Glenn Palmer.

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Causes of Escalation

Our intent is to explore the implications of how escalation is measured. By doing so, we can observe how consistently a number of suspected covariates perform across different measures of escalation. In this section, we identify a selection of variables that past researchers have theorized and/or shown empirically affect conflict escalation. Accordingly, the current article is meant to capture what we see as the most widely accepted factors affecting the probability of escalation, but it is, clearly, not an exhaustive review of all arguments about escalatory processes or the variables that have been found to affect escalation in some way. By focusing our statistical work on this influential subset of independent variables, we hope to see how widely generalizable across operational definitions of "escalation" the range of seminal findings are.

First, there is a strong tradition of research into patterns of escalation within the democratic peace literature. Maoz and Abdolali (1989) were the first to note that while states with democratic political systems do come into conflict with each other, their conflicts do not escalate to war. Maoz and Russett (1993) use a dispute's highest action on the MID hostility scale to indicate escalation (see also Hart and Reed, 1999) and show that joint democracy reduces the likelihood of escalation. Following suit, Senese (1997) finds that the pacifying effect associated with joint democracy is more pronounced at lower rather than higher levels of conflict. Separately, Prins and Sprecher (1999) have shown that democratic targets are more likely to reciprocate MIDs when their opponent is a non-democracy, and Prins (2001) has demonstrated that the United States specifically is more likely to reciprocate a dispute if the initiator is a non-democracy.

A series of studies about the effects of domestic politics upon conflict processes further refine and disaggregate their operationalizations of escalation. In testing the domestic variant of their international interactions game, Bueno de Mesquita and Lalman (1992) use the hostility levels reached by the two sides in a dispute to indicate various alternative outcomes. They distinguish between mutual use of force versus being involved in a COW war. Schultz's (2001) model of signaling behaviors prompts several hypotheses pertaining to the influence of domestic constituents upon various forms of conflict escalation. He uses three measures to capture these: the reciprocation of force, the mutual use of force, and war (Schultz, 2001: 263–264). In testing their hypotheses about right versus left parliamentary governments, Palmer et al. (2004) use two indicators of "escalation": disputes that have more than 25 battle fatalities, and the reciprocation of a dispute at a level of hostility at least as high as the initiator's first action. This latter indicator is also employed by Partell (1997) in his test of Fearon's domestic politics model of dispute escalation. Across each of the models mentioned here, democracy is shown to have a constraining effect upon the escalation of hostilities between states.

A second factor associated with escalation is the issue at stake in a conflict. Gochman and Leng (1983) and Leng and Singer (1988) find that combatants are more likely to escalate to war when territorial integrity is at stake. Senese (1996) demonstrates that contiguity and the presence of territorial disagreements do not increase the observed levels of hostility but that once force is used in territorial conflicts, greater fatalities occur. Similarly, Ghosn et al. (2004) find that disputes

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over territory are more likely to involve fatalities and higher levels of fatalities than is the case when territorial issues are not present. Finally, Hensel (1996) shows that disputes are more likely to be reciprocated and to escalate to war if they involve territorial issues—this second finding is also confirmed by Vasquez and Henehan (2001). So far as this represents a fair summary of the literature, it highlights strong evidence of a positive relationship between competition over territory and the escalation of conflict.

Third, drawing upon the arguments of power transition theory (Organski and Kugler, 1980; Lemke and Reed, 1996; Lemke, 2002), a number of scholars argue that satisfaction with the status quo affects the onset and escalation phases of conflicts. According to this logic, states satisfied with current international arrangements have less incentive to fight each other than do dissatisfied states. On a related matter, a fourth set of studies address the relationship between power preponderance and conflict processes. Some claim that preponderance increases the likelihood of war (Waltz, 1979), while others maintain that preponderance decreases the likelihood of both conflict (Kugler and Lemke, 1996) and escalation to war (Moul, 1988). Fifth, numerous studies investigate the impact of joint alliance membership on the likelihood of conflict and its escalation (Bueno de Mesquita, 1981; Bremer, 1992). The evidence on the effect of joint alliance membership is mixed.

Our summary of the existing literature highlights five variables (joint democracy, territorial disagreement, status quo satisfaction, power preponderance, and alliance ties) that represent the most commonly expected correlates of escalation. We investigate how robust these hypothesized correlates prove to be when we vary how escalation is measured. Toward that end, we next introduce our six measures of conflict escalation.

Measures of Escalation

The studies described above mostly use information from the Militarized Interstate Dispute (MID) dataset to indicate escalation. In all they describe five simply measured, alternate predictors of escalation. We generalize across these studies and use as measures of escalation different subsets of the overall population of Militarized Interstate Disputes. By focusing on escalation once disputes have begun, we cannot offer escalation measures independent of the existence of MIDs. But since the majority of escalation research studies MIDs, this restriction is reasonable.[1]

Our first measure of escalation indicates whether the dispute is "Reciprocated", that is, whether the target of the dispute responds in kind or at some higher level of militarized action. Of the full set of 1901 MIDs[2] (1816–2001), 917 (48% of

[1] For studies that consider escalation from verbal hostilities to disputes, or from crises to violent clashes, see Kinsella and Russett (2002) and Colaresi and Thompson (2002). In future work it might be instructive to create similar measures of escalation for ICB crises and to compare 25 and 1000-fatality ACD conflicts.

[2] This total is fewer than the 2331 total MIDs in the MID3.0 dataset, because we have omitted all MIDs that began as multilateral disputes and have focused only upon those that were bilateral on day one.

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the total) were reciprocated—that is, escalated according to this measure. Our second measure of escalation records whether a "Use of Force" occurred during the dispute—measured using the MID dataset's hostility variable. According to this measure, 1394 (73%) of MIDs escalated. Our third measure of escalation combines the first two. A "Mutual Use of Force" takes a value of 1 if both sides used force during the dispute. This was the case in 576 (30%) of the MIDs. Our final three measures of escalation reflect increasing levels of fatalities in MIDs. "Fatalities > 0" takes a value of 1 only in disputes where at least one combatant died. This occurred in 536 (28%) of the MIDs. "Fatalities > 250" indicates more substantial loss-of-life situations and takes a value of 1 only if at least 250 soldiers died during the MID. This occurred in 294 (15%) of the MIDs.

Our final measure of escalation indicates whether the dispute reached the 1000+ battle fatalities threshold qualifying the dispute as an interstate war. The "War" escalation measure takes a value of 1 in 104 (5%) of the MIDs. War is the most commonly used measure of escalation in past empirical studies, starting with Wallace's arms race studies (1979, see also Diehl, 1983) and continuing as recently as Senese and Vasquez's study of the steps-to-war (2008). Scholars have evaluated hypotheses against different measures of escalation, as indicated above, but the bulk of past studies have used escalation of MIDs to war as their dependent variable. Our study is unique because we include five other measures of escalation.

Pairwise correlations between the six escalation measures vary widely, from a high of 0.756 (between "Mutual Use of Force" and "Fatalities > 0") to a low of 0.144 (between "Use of Force" and "War"; all correlations are statistically significant). That the various operationalizations are only marginally related to each other empirically suggests that there may be distinct types of escalation.

Analyses of Escalation

We employ each of the escalation measures just described as dependent variables and regress them against each of the five independent variables highlighted by previous research. We employ a comprehensive dataset of non-directed dyads for the years 1919–1995. All told, there are just under 500,000 observations, among which 1352 bilateral MIDs[3] occurred. Our priority is to offer an analysis of the causes of escalation; but in order to estimate the influence of the five variables on escalation we must also determine their influence on the onset of the MIDs in the first place. Early studies of escalation paid little attention to problems associated with selection bias—the notion that since states select themselves into conflict, analyses of escalation restricted to conflicting states do not employ a random or representative sample. Statistical analyses that focus only on escalation, without regard to the prior stage of conflict onset, likely produce biased results. Reed (2000) offers one of the first attempts to unite analyses of stages of conflict (see also Gartner and Siverson, 1996; Hart and Reed, 1999; and Lemke and Reed, 2001). He operationalizes the first stage of his model as dyadic involvement in a MID and the second stage as

[3] This number is, again, fewer than the 1901 MIDs discussed earlier because of the need to truncate our temporal domain due to data availability on our covariates.

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the escalation to war, showing that some factors (such as joint satisfaction and joint democracy) produce smaller effects when a two-stage analysis is used. In the present study we employ the same Heckman-style procedure. This permits us to determine whether the presence or absence of selection bias is uniform across the six escalation measures and whether the presence of that bias has substantive implications for the generalizability of the results.

The dependent variable in the model's first stage is MID Onset. For each of the nearly 500,000 observations this variable equals 1 only in dyad years in which a dispute began. Data for MID onsets are described by Ghosn et al. (2004). The second dependent variable, Escalation, is recorded only for those 1352 observations in which a MID occurred. Dyad years without MIDs are censored and excluded from calculation of the independent variables' influences on escalation. Information on escalation is also contained in Ghosn et al. (2004).

The independent variables justified above are included in estimation of both dependent variables. In addition, three other covariates are included in analysis of MID Onset. Their inclusion is motivated by statistical necessity. Multi-stage models, like the censored probit used here, require predictors to identify the first stage. It is particularly advantageous if the variables chosen are strongly correlated with the first stage dependent variable and uncorrelated with the second. As such, their inclusion is predicated more on model fit than on drawing inferences. The predictors we use to identify the first stage of MID Onset are Contiguity, Rivalry, and Minor-Minor Status. These three variables are excluded from estimation of MID escalation.

Joint Democracy draws on information available in the Polity IV dataset (Jaggers and Gurr, 1995). It is a dichotomous variable, equal to 1 if both dyad members have democracy scores of six or greater. While there is substantial agreement that we should expect a negative coefficient for Joint Democracy's influence on MID Onset, there is disagreement about the expectation regarding Joint Democracy's subsequent influence on Escalation. A straightforward expectation would be that Joint Democracy will decrease the probability of Escalation, because democracies are alleged to favor nonviolent dispute resolution. On the other hand, it may be that Joint Democracy's pacifying influence is so strong at the MID Onset stage that if any jointly democratic dyads experience a MID Onset in spite of their regime types, that dispute would likely be so severe that even the democracies could not resolve it peacefully. If that were true, then the expectation might be that Joint Democracy would be negatively associated with MID Onset but positively associated with Escalation (a variation on this expectation is that Joint Democracy should be expected to have no relationship with escalation at all, controlling for its prior influence on MID Onset; for a discussion of such expectations, see Kinsella and Russett, 2002).

Our second expected correlate of escalation is Territorial Dispute. The MID dataset includes information about the issue central to each dispute. Territorial Dispute is a dichotomous variable equal to 1 whenever the MID that occurred specifically involved a disagreement about territory. Since territorial disagreements are widely reported to be disproportionately difficult to resolve peacefully, the expectation is that Territorial Dispute will be positively associated with escalation.

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In estimating the influence of territorial issues on MID Onset, we obviously cannot use data about whether a MID was a dispute over territory as we would lack data on all the Onset = 0 cases. Instead, we employ Huth and Allee's (2002) dataset indicating states with Territorial Claims between them. We use this variable to represent territorial issues as a predictor of MID onset. Huth and Allee's data are only available for the period 1919–1995, necessitating the time period studied here.

Joint satisfaction is a dummy variable indicating whether both dyad members have positive tau-b alliance portfolio similarity scores with the international system's leading state (UK 1919–1945, USA 1946–1995). Based on power transition theory, the expectation is that dyads of jointly satisfied states have such affinity that they will avoid conflict with each other and be likely to find negotiated settlements for any disputes that do arise between them. Conventional wisdom leads us to expect negative coefficients for Joint Satisfaction.

We measure Power Preponderance with COW's composite indicator of national capabilities (CINC) score (first described by Singer et al., 1972). We use it to calculate the share of dyadic capabilities possessed by the stronger member of the dyad. Thus, it runs from 0.5 (perfect equality within the dyad) to 1 (the stronger state is completely preponderant). We include power preponderance in estimation of both dependent variables, and if the conventional wisdom is correct we will uncover negative estimates for its effect in both stages—with preponderance being associated with the absence of both conflict and escalation.

Defense Pact indicates whether members of the dyad are united in a treaty of alliance that provides security guarantees of mutual assistance in the event either is attacked. Defense Pacts are widely regarded as the alliance type indicating the highest degree of common security interests, and thus it is specifically this type of alliance relationship that should correspond with MID avoidance and non-escalation. Past research leads us to anticipate negative coefficients for Defense Pact in both stages of our censored probit analyses of escalation. Data about defense pacts are made available by Gibler and Sarkees (2004).

Our unique predictors of MID Onset are also drawn from common sources. Contiguity is a dichotomous variable equal to 1 if the dyad members are directly contiguous or separated by fewer than 125km of water (Stinnett et al., 2002). This variable should be positively associated with MID Onset. Similarly, Rivalry is a dummy variable representing whether the dyad members identify each other as significant threats. Drawing on Thompson's (2001) rivalry data, this variable equals 1 for any year in which a rivalry existed between dyad members. Finally, Minor-Minor Status is another dichotomous variable equal to 1 when both dyad members are Minor Powers. The COW project provides a list of Major Powers (Small and Singer, 1982: 44–46), and all states omitted from that list are recorded as minor powers. We expect that pairs of minor powers are less likely than major powers to escalate disputes. These three variables correlate highly with MID Onset, but have inconsistent (sometimes positive, sometimes negative) and only modest correlations with the escalation measures. Finally, to control for duration dependence, we have included Carter and Signorino's (2010) cubic polynomial specification of Peace Years.

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Table 1. Selection Models of Dispute Onset and Escalation

	Reciprocated	*Use of Force*	*Mutual use of Force*	*Fatalities > 0*	*Fatalities > 250*	*War*
Onset						
Contiguity	1.06***	1.06***	1.06***	1.06***	1.06***	1.06***
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
Rivalry	0.80***	0.80***	0.80***	0.80***	0.79***	0.79***
	(0.09)	(0.09)	(0.09)	(0.09)	(0.09)	(0.09)
Both Minors	−0.71***	−0.71***	−0.70***	−0.71***	−0.71***	−0.72***
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Joint	−0.21***	−0.21***	−0.21***	−0.21***	−0.21***	−0.21***
Democracy	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Joint	−0.001	−0.003	−0.001	−0.001	−0.003	−0.004
Satisfaction	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)
Territorial	0.50***	0.51***	0.50***	0.50***	0.50***	0.51***
Claim	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)
Power	−0.27*	−0.29*	−0.27*	−0.27*	−0.29*	−0.29*
Pre-ponderance	(0.16)	(0.16)	(0.16)	(0.16)	(0.16)	(0.16)
Allied	−0.003	−0.003	−0.004	−0.004	−0.003	−0.002
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
Peace	−0.03***	−0.03***	−0.03***	−0.03***	−0.03***	−0.03***
Years	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Constant	−2.05***	−2.04***	−2.07***	−2.06***	−2.04***	−2.03***
Escalation						
Joint	0.01	−0.06	0.25*	0.17	0.16	0.36**
Democracy	(0.13)	(0.14)	(0.13)	(0.13)	(0.15)	(0.18)
Territorial	0.61***	−0.13	0.50***	0.46***	0.25**	0.44***
Dispute	(0.11)	(0.10)	(0.11)	(0.11)	(0.13)	(0.14)
Joint	−0.21*	−0.26*	−0.36**	−0.58***	−0.37**	0.15
Satisfaction	(0.14)	(0.14)	(0.15)	(0.15)	(0.16)	(0.20)
Power	−0.26	−0.04	−0.04	0.08	0.63**	0.47
Pre-ponderance	(0.28)	(0.30)	(0.32)	(0.30)	(0.32)	(0.44)
Allied	0.04	−0.04	0.07	0.16	0.08	−0.33
	(0.13)	(0.13)	(0.13)	(0.13)	(0.14)	(0.23)
Constant	0.72***	0.73***	0.06	−0.05	−1.08***	−1.75***
Rho	−0.33***	0.07	−0.30***	−0.30***	−0.23***	−0.26***
Stage 1 N	499,145	499,145	499,145	499,145	499,145	499,145
Stage 2 N	1352	1352	1352	1352	1352	1352
Model Chi^2	35.97***	9.32*	28.32***	27.30***	12.18**	17.25***

Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$. Cell entries are coefficients with robust standard errors clustered on the dyad in parentheses. Squared and cubic Peace Years coefficients not reported to save space.

Table 1 reports censored probit estimates and summary statistics from six selection models of the escalation measures against all five alleged correlates of escalation, controlling for their prior influence on the onset of MIDs. With respect to MID Onset, expectations based on past studies are almost uniformly supported. Only Joint Satisfaction and Alliance fail to perform as expected in the upper half of Table 1.

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Table 2. Predicted Probabilities of Escalation Types

	Reciprocated	*Force*	*Mutual Use of Force*	*Fatalities > 0*	*Fatalities > 250*	*War*
Underlying Probability	0.48	0.77	0.32	0.31	0.158	0.041
Democracy = 0			0.31			0.038
Democracy = 1			0.342			0.059
Joint Satisfaction = 0	0.4928	0.77	0.3264	0.3247	0.1630	
Joint Satisfaction = 1	0.4635	0.7623	0.2752	0.2257	0.1124	
Territorial Dispute = 0	0.4489		0.2816	0.269	0.1377	0.0311
Territorial Dispute = 1	0.5611		0.3872	0.371	0.1789	0.0588
Preponderance = Mean − 1 s.d.					0.1219	
Preponderance = Mean					0.1583	
Preponderance = Mean + 1 s.d.					0.1836	
Defense Pact = 0						0.0447
Defense Pact = 1						0.0311

Cell entries are predicted probabilities of the column outcome given the row condition, holding other variables at means or modes. The underlying probability of each form of escalation, given that a dispute has occurred and holding all covariates at their means, is provided in the top row. Empty cells indicate the row variable is not a significant predictor of that type of escalation.

Turning to the lower half of the Table 1, which reports the estimates of the impact of the five explanatory variables on the likelihood of MID escalation, the results are far less supportive of the conventional wisdom. Power Preponderance and Defense Pact are only rarely significantly associated with Escalation, regardless of how escalation is measured. Similarly, Joint Democracy makes escalation *more* likely in the only instances where it achieves statistical significance. Joint Satisfaction is negatively related to Escalation in five of the six models, in contrast to its non-performance predicting MID Onset. Finally, we find that the Territorial Dispute variable's influence is clearly the strongest of those included in the second stage. In five of six estimations this variable significantly increases the probability of escalation. Only in instances of unilateral uses of force do we find no statistically significant effect of territory upon escalation.

Table 2 presents substantive effects of the five covariates on the six "types" of escalation. The table reports predicted probabilities. The substantive story told by the predicted probabilities in Table 2 mirrors that told by the coefficient estimates in Table 1. In almost every instance, the substantive effect of Territorial Dispute is far larger than that of any of the other variables. Territorial Dispute is thus both the most significant and substantively important predictor of escalation across all types. Also of interest in Table 2 is the observation that the marginal effect of significant covariates generally increases as we move from the left to the right side

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of the table. Arguably this means that the "usual suspects" of escalation are more important predictors of more violent forms of escalation.[4]

Turning to more theoretically substantive interpretations of the results of our censored probit estimations, our results offer insights about escalation and the democratic peace, power transition theory, and the territorial explanation for war.

The straightforward Democratic Peace expectation that joint democracy will make both onset and escalation less likely is roundly discredited by our analyses. While joint democracy consistently makes dispute onset less likely, it never makes escalation less likely. The expectation that democratic dyads finding themselves in disputes must be so strongly in disagreement that escalation is *more* likely, appears to be supported. However, the positive effect of Joint Democracy on escalation to War is based on the existence of eight jointly democratic warring dyads. Chronologically they are: (1&2) both France and Greece v. Turkey (1919), (3) Lithuania v. Poland (1920), (4) Britain v. Finland (1941), (5) Hungary v. Romania (1944), (6&7) both Cyprus and Greece v. Turkey (1974), and (8) India v. Pakistan (1993—this MID escalates to war in 1999). Two of these pairings are odd artefacts of the way that our datasets typically code World War II. Two further pairings reflect the problem of regime type being measured at the beginning of the year, which does not reflect that coups or other regime transitions occur during the course of the year. As a consequence, it is conceivable that transitions are occurring after a war rather than before. If we remove the four objectionable "democratic wars" (4, 5, 6 and 7), Joint Democracy is no longer significantly related to escalation to war. For escalation to Mutual Use of Force, there are 43 jointly democratic dyads and fewer clear "outliers" to remove. The escalation-enhancing effect of Joint Democracy on mutual use of force seems reasonably valid (consistent with Senese, 1997). A summary statement about the Democratic Peace is that it seems to tell a more convincing story about conflict onset than it does about escalation.

Power transition theorists might claim that Joint Satisfaction performs nearly as well as Territorial MID in the Escalation phase of the censored probits. But enthusiasm for Joint Satisfaction as the best predictor of Escalation must be tempered by the variable's contrary performance predicting the initial onset of disputes. The power transition waters are further muddied if we specify Joint Satisfaction with Signorino and Ritter's (1999) S measure of affinity for the dominant state. With Joint Satisfaction measured by S, Joint Satisfaction is uniformly pacifying at the onset stage and uniformly insignificant at the escalation stage. A final point of concern for power transition explanations of escalation is found in the inconsistent effect of preponderance across measures of escalation. Power transition anticipates preponderance to be conflict dampening, and thus the Preponderance coefficient should be negative. For Onset it uniformly is. For escalation, it is negative but insignificant at lower levels of escalatory severity.

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[4] The predicted probabilities for Defense Pact on "War" are included even though the coefficient on Defense Pact is not statistically significant. With a one-tailed test Defense Pact would be significant at the $p < 0.10$ level. Readers may ignore or interpret the modest pacifying influence of Defense Pact on Escalation to war as they like.

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However, once fatalities define escalation, preponderance makes escalation *more* likely. In the case of Fatalities > 250, this effect is statistically significant. Like Democratic Peace proponents, Power Transition theorists might conclude from these results that their theory has more to say about onset than escalation. How that modifies their theory, future work may show.

Proponents of the territorial explanation for war will be very encouraged by these findings. We think they are right to be so, and an obvious next step for them would be to continue developing evidence about whether causal links from Territoriality to Escalation really exist. Steps-to-War theory places territorial disagreements at the beginning of a causal story in which leaders have been conditioned to react to territorial disagreements with militarized, realpolitik tools. These make conflict harder to resolve, and war follows. There is some strong evidence supporting Steps-to-War theory (Senese and Vasquez, 2003, 2008; Colaresi and Thompson, 2005). More explicit evidence developing the links from Territory to Policies to Escalation would further increase confidence in the theory.

Conclusions

We conclude with the relatively straightforward message that research on escalation needs to consider carefully how the dependent variable is measured. Past research suggests a set of five likely causes of escalation. Our sensitivity analysis across escalation types demonstrates that only territorial disputes consistently escalate.

We uncovered considerable evidence supporting claims that analysis of escalation cannot be separated from analysis of conflict onset. In all but one of the censored probit models the selection parameter, *rho*, was statistically significant and negative. To an extent rarely seen in other applications of this statistical modeling technique, we uncover persistent results that consistently suggest Onset and Escalation are linked parts of a single process. This suggests that the use of such estimators is appropriate in the analysis of escalation.

In terms of theory about escalation, two main points follow. First, not all forms of escalation appear to be the same. Specifically, the model employing "Use of Force" escalation as the dependent variable produced the fewest significant results. There appears to be something different about escalation instances where a MID achieves "Use of Force" status, than when it experiences escalation in some other fashion. What causal process might distinguish this form of escalation from the others? Might Use of Force MIDs be least likely to be reciprocated? Might they be pre-emptive or preventive strikes? We have demonstrated that this one type of escalation persistently performs differently than the others but do not know why.[5]

[5] In unreported analyses we also measured Escalation as occurring when an initially bilateral MID was transformed into a multilateral one through the actions of a third-party joiner. Clearly, joiners increase their hostility levels simply by joining the dispute. But in these "Bilateral-to-Multilateral" Escalation analyses there is no significant predictor of escalation, and none of these models was jointly significant. This means that in addition to "Use of Force" there are other, different types of escalation that might motivate different causal stories. Similarly, another form of escalation omitted here is the geographic spread of MIDs as they expand to involve conflict over larger sections of the globe. Data about the geographic

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The second general point for subsequent theory development is based on the fact that, with the exception of Territorial Dispute, there really is no good predictor of Escalation (regardless of how we measure Escalation) within the corpus of existing empirical research on Escalation. Our conventional wisdom and past empirical work has failed to produce any other "smoking gun" causes of Escalation.

Perhaps this is because we have been looking too far removed from the actual behaviors occurring within MIDs. Whether a disputing dyad is unequal in power, or allied, or jointly democratic, are all characteristics known ahead of time. As such it is quite reasonable to find that they influence MID Onset. But once the MID has begun, is it still reasonable to expect that those fixed characteristics can influence the changing nature of the unfolding MID? Formal theorists certainly suggest it is not (Morrow, 1989; Bremer and Cusack, 1995; Bueno de Mesquita et al., 1997). Perhaps we must look for new data sources, about the actual behaviors undertaken by disputing states, to find powerful predictors of escalation. Of course, to the extent that such behaviors can be anticipated by putative disputants before the MID begins, we are back where we began with the fixed, pre-existing data. It is possible that Steps-to-War researchers gathering data on realpolitik policies within disputes will provide the information we need. A different, but logically related, effort to use changing conditions within MIDs (such as the modification of the pre-MID distribution of capabilities a third-party joiner causes) are promising new directions (Bennett, 2008). Also potentially useful is the observation that early studies of the democratic peace focused on democracies' abilities to resolve their disagreements via mediation (e.g. Dixon, 1993). Perhaps established democracies are better able to secure effective mediators than are new democracies. Maybe development along these lines could explain why democracy has such inconsistent effects on escalation.

Those are speculations about future work. Our surest conclusion based on the analyses reported here is that, of all the known and suspected correlates of escalation, only the presence of territorial issues at stake is a consistently good predictor of whether disputes will escalate.

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spread of MIDs are only available after 1991, and this restriction motivates their exclusion from our analyses. But readers interested in geographic spread as escalation should consult Braithwaite (2006, 2010).

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